

AI USAGE REPORT

Course: CSC13002 – Introduction to Software Engineering

Project: LifeLine – Blood Donation Management Platform

Team: Sanguine (Group 05)

Report Phase	Sprint 2 – PA3
Submission Date	25/07/2026

A. Declaration of AI Use

- ☐ A. Declaration of No AI Use — "Throughout the entire project/assignment, the student did not use any AI tools in any form."
- ☒ B. AI Usage Notes — The AI tools (Claude, ChatGPT, Gemini, Google Stitch, and Antigravity IDE with Speckit) were used as assistants for drafting specifications, generating UI prototypes, and supporting the coding implementation through Spec-Driven Development (Speckit). Full details are provided in Section B below. Every project decision (problem framing, scope, workflows, business logic, and architecture) was made independently by the team.

B. AI Usage Notes

B.1. Trần Anh Kiệt – Project Manager

Session 1: Use-Case Specification — User Account Management & Donation Booking (FG 1.1 & 1.2)

Field	Content
Tool name	Claude (Anthropic)

Field	Content
Version	<i>Sonnet 3.5 High</i>
Platform	claude.ai (Web)
Access time	21/06/2026, 19:11
Purpose of use	Draft the Use-Case Specification (UCSPEC) markdown for Functional Groups 1.1 "User Account Management" and 1.2 "Donation Booking & Location Services," per the "D – Use-Case Specification" requirement in PA3-2026.pdf, using VisionDocument.pdf as the source of truth and following the format of a reference document from another team (LC_UseCase_v.1.5.pdf) and a teammate's existing UC file.
Prompts used	1. Bây giờ hãy đọc thật kĩ file PA3-2026.pdf đặc biệt là phần "D - Use-Case Specification", Hãy đọc file VisionDocument.pdf như là source of truth và dựa vào nó để hiểu context của đề án nhóm chúng tôi. Đặc biệt là đọc kĩ phần 5. Product Features. Bây giờ tôi đã gửi kèm theo đó là 2 file tham khảo 1 là file "LC_UseCase_v.1.5.pdf", đây là file viết Use Case Spec của 1 nhóm đã làm trước, hãy tham khảo cách viết của họ, còn file "UC (1).md" là file của 1 thành viên trong nhóm của tôi đã làm về các phần "2.1 Campaign & Donor Mgmt, 2.2 Communication & Engagement, 2.3 Blood Inventory, 4.1 User Automations + 4.2 BC Automations" trong vision doc. Bây giờ hãy làm lại cho tôi file UseCaseSpec của 1.1 User Account Mgmt & 1.2 Donation Booking & Location tương ứng với phần 5.1.1. User Account Management [Đính kèm: UC (1).md, LC_UseCase_v.1.5.pdf, VisionDocument.pdf, PA3-2026.pdf]
Which content was generated by AI	UC_UserAccount_DonationBooking.md , covering LL-UC-01 through LL-UC-10 (User Account Management + Donation Booking & Location Services), including a Mermaid flowchart diagram and full use-case tables (Basic Flow, Alternative Flows in AF-0x convention, Postconditions, Special Requirements, Related Use Cases).
Independent Content & Validation	The student read through the whole file after it was generated and checked it against VisionDocument.pdf and PA3-2026.pdf to make sure nothing from section 5.1.1 was missing then compared the format with the teammate's UC (1).md file to keep the numbering and table style the same across the team, and fixed a few small wording spots so the flows matched how the booking feature is actually supposed to work in the app.

Session 2: Use-Case Specification — System Features (Backend & Automations, FG 4.1 & 4.2)

Field	Content
Tool name	Claude (Anthropic)
Version	Sonnet 3.5 High
Platform	claude.ai (Web)
Access time	21/06/2026, 22:37 – 23/06/2026, 08:02
Purpose of use	Draft the Use-Case Specification for system-level features (5.4 System Features in the Vision Document), covering SYS-UC-01 through SYS-UC-04 across sub-groups 4.1 "User-Facing Automations" and 4.2 "Blood Center-Facing Automations"; later correct the content of SYS-UC-03 after the student flagged a conceptual error.
Prompts used	1. Tương tự như trên, hãy làm cho tôi file markdown Use Case Spec của 5.4 System Features (Backend & Automations) trong vision doc, Dựa vào các file trên tôi gửi bao gồm "UC (1).md", VisionDocument.pdf, "LC_UseCase_v.1.5.pdf", UseCaseSpec1.1&1.2.pdf [Đính kèm: UseCaseSpec1.1&1.2.pdf] 2. Trong SYS-UC-03: Generate Digital Donor Record, nội dung bị lệch so với ý tưởng ban đầu, thứ nhất đây là use case dành cho blood center staff, và công dụng của nó là để nhân viên lấy máu thực hiện trong quá trình hiến máu của user... NỘI DUNG UC3 NÀY HIỆN TẠI LÀ nó đang là dùng để lưu thông tin sau khi hiến máu, và nó được gen sau khi quá trình hiến máu xảy ra là sai, hãy sửa lại nội dung theo những gì tôi đã đề cập trên
Which content was generated by AI	UC_SystemFeatures.md (SYS-UC-01 to SYS-UC-04, overview table + Mermaid diagram); corrected version of SYS-UC-03 (reframed from a post-donation record to a registration record auto-created at booking time and used live by staff during the donation day), with corresponding updates to the Overview table and to SYS-UC-02's Related Use Cases field.
Independent Content & Validation	The student identified and corrected the conceptual error in SYS-UC-03 (timing/purpose of the digital donor record) based on the team's actual workflow understanding. After that the student also went through SYS-UC-01, SYS-UC-02 and SYS-UC-04 himself to make sure the overview table and the

Field	Content
	Related Use Cases field of SYS-UC-02 were updated correctly to match the fixed version of SYS-UC-03.

Session 3: Use-Case Specification — Hospital Features (Related Use Cases completion)

Field	Content
Tool name	Claude (Anthropic)
Version	<i>Sonnet 3.5 High</i>
Platform	claude.ai (Web)
Access time	23/06/2026, 08:35
Purpose of use	Fill in the empty "Related Use Cases" fields in the Hospital Features UCSPEC file, mapping include, extend, triggered-by, updated-by, and data-sourced-from relationships among the four hospital use cases.
Prompts used	1. Bây giờ file UCSpec-Hospital.md này đang bị thiếu các related Use case, hãy bổ sung vô cho tôi các related UC giữa những thằng UC Spec này thôi [Đính kèm: UCSpec-Hospital.md]
Which content was generated by AI	Completed "Related Use Cases" fields for HS-UC-01, NS-UC-02, HS-UC-03, and HS-UC-04 in <code>UCSpec-Hospital.md</code> .
Independent Content & Validation	I checked each of the four hospital use cases one by one to see if the include, extend or trigger relationships that Claude filled in actually matched what was already written in the Basic Flow and Description of each use case, then kept the ones that made sense and left a note to double check HS-UC-04 later since that one was still being changed by another teammate.

Session 4: Use-Case Specification — Consistency Review of "Related Use Cases" (Full Document, Attempt 1)

Field	Content
Tool name	Claude (Anthropic)

Field	Content
Version	<i>Sonnet 3.5 High</i>
Platform	claude.ai (Web)
Access time	23/06/2026, 08:35
Purpose of use	Review the full <code>UseCaseSpec.md</code> (written collaboratively by multiple teammates, each responsible for different use cases) to check whether the "Related Use Cases" field of every use case correctly and completely reflected UML include, extend, and generalization relationships.
Prompts used	1. Đây là file UseCaseSpec.md của tôi, hiện tại tôi chia việc cho từng người làm các use case nên là tổng thể nó sẽ rời rạc và ko liên kết nhau, hãy đọc thật kĩ file và phân tích xem các use case đã thống nhất và liên quan với nhau chưa, hãy kiểm tra xem ở phần related use cases của mỗi use case đã đầy đủ và chính xác chưa (nếu chưa đầy đủ hoặc chưa chính xác, hãy viết lại cho tôi phần related use case của mỗi use case sử dụng các mối quan hệ include, extend, generalization trong UML). [Đính kèm: UseCaseSpec.md , VisionDocument.pdf]
Which content was generated by AI	<i>No output produced</i> — Claude was unable to load the file content across repeated attempts (technical/tool issue), and the review was not completed in this session.
Independent Content & Validation	Since Claude could not load the file properly in this session and nothing was actually produced, there was nothing here for me to edit or check. I tried resending the file a couple of times before giving up and moving the review to a later session instead.

Session 5: Use-Case Model — Full Mermaid Diagram Set (all actor groups)

Field	Content
Tool name	Claude (Anthropic)
Version	<i>Sonnet 3.5 High</i>
Platform	claude.ai (Web)
Access time	23/06/2026, 13:01

Field	Content
Purpose of use	Generate a complete Use-Case Model in Mermaid format (embedded in Markdown) covering all functional requirements, organized by actor group: Donor/User (1.1–1.6), Blood Center Staff (2.1–2.3), Hospital Staff (3.1), System automation (4.1–4.2), and Administrator (5.1), using <code>UseCaseSpec.md</code> as source of truth and a reference image for visual style.
Prompts used	1. Bây giờ hãy đọc và ghi nhớ thật kĩ file UseCaseSpec.md này, hãy coi là source of truth và viết cho tôi Sơ đồ UseCase Model cũng tương tự như cái ảnh tôi đã gửi, nhưng làm thành diagram mermaid trong markdown... [Chi tiết yêu cầu về các actor nhóm chức năng] [Đính kèm: UseCaseSpec.md , ảnh sơ đồ mẫu]
Which content was generated by AI	<code>UseCaseModel.md</code> — six Mermaid diagrams (five actor-specific diagrams plus one full system overview diagram) covering 52+ use cases, with «include» and «extend» relationships and supporting system actors (Notification Engine, Notification Service, System automated) modeled with dashed association lines.
Independent Content & Validation	I opened my own UC Diagram and checked each of the six diagrams one at a time against UseCaseSpec.md to make sure every use case from every actor group was actually shown and that the include and extend arrows pointed the right way, and compared the overall look with the reference image the team was given before accepting it.

Session 6: UI/UX Design for User Features Based on UseCaseSpec (Google Stitch prompts)

Field	Content
Tool name	Claude (Anthropic)
Version	<i>Sonnet 3.5 High</i>
Platform	claude.ai (Web)
Access time	27/06/2026, 12:56 – 30/06/2026, 21:08
Purpose of use	Using section 2.1 "User Features" of <code>UseCaseSpec.md</code> as source, write detailed prompts for Google Stitch to generate UI screens for donor-facing use cases (Register, Login, Logout, Reset Password, Manage Profile,

Field	Content
	Schedule Appointment) and their alternative flows, matching an existing Blood Center Staff screen template.
Prompts used	1. Bây giờ tôi có cái UseCaseSpec.md như này, hiện tại tôi cần phải làm giao diện cho UseCase của User Features bao gồm toàn bộ mục 2.1... viết cho tôi prompt để đưa lên Google stitch để nó thiết kế giao diện cho tôi. [Đính kèm: UseCaseSpec.md, DESIGN.md, ảnh giao diện mẫu] 2. [Prompts bổ sung về template, thống nhất logo/avatar/footer và các Alternative Flows biến thể trên cùng màn hình]
Which content was generated by AI	Google Stitch UI-generation prompts for the Register, Login, Logout, Reset Password, Manage Profile, and Schedule Appointment screens (including their AF variants) and a screen-consistency standardization prompt.
Independent Content & Validation	The student supplied the actual Basic Flow / Alternative Flow text for each use case from the team's UseCaseSpec, and directed the correction that AF screens must be shown as variations of the same screen rather than separate pages. The student then ran the prompts in Google Stitch, reviewed and tweaked the results.

Session 7: Use-Case Relationship Review — Full Specification (Attempt 2, partial)

Field	Content
Tool name	Claude (Anthropic)
Version	<i>Sonnet 3.5 High</i>
Platform	claude.ai (Web)
Access time	08/07/2026, 21:58
Purpose of use	Re-attempt a full audit of <code>UseCaseSpec (2).md</code> 's "2. Use-case Specifications" section, checking whether the "Related Use Cases" field of each use case correctly reflects include/extend/generalization relationships.
Prompts used	1. Đây chính là file UseCaseSpec (<code>UseCaseSpec (2).md</code>) của nhóm tôi... hãy xem các mối quan hệ Related Use Cases của nó hiện đã thể hiện đúng các mối quan hệ include, extend, generalization trong UseCase diagram hay không. [Đính kèm: <code>UseCaseSpec (2).md</code>, vision.md, <code>PA3-2026.pdf</code>]

Field	Content
Which content was generated by AI	Partial review: identified reversed extend directions, missing extend relationships, and fabricated relationships; review of Hospital, System, and Administrator sections was not completed in this session.
Independent Content & Validation	This session did not really finish since Claude only got through part of the file before the review stopped, so the student just kept the partial findings as a reference and did not act on them yet.

Session 8: Use-Case Relationship Review — Full Specification (Attempt 3, completed, with corrections)

Field	Content
Tool name	Claude (Anthropic)
Version	<i>Sonnet 3.5 High</i>
Platform	claude.ai (Web)
Access time	08/07/2026, 20:20 – 21:08
Purpose of use	Complete the pairwise consistency audit of all 55 use cases' "Related Use Cases" fields across all actor modules against the documented flows, then apply the student's proposed corrections.
Prompts used	1. [Yêu cầu kiểm tra Related Use Cases, không được hallucinate] 2. [Thảo luận về cơ chế SOS-UC-01 và SYS-UC-05] 3. [Đề xuất 5 chỉnh sửa cụ thể cho module Notifications và Admin]
Which content was generated by AI	Full audit results: identified missing relationships, flows lacking support, and minor textual inconsistencies; confirmation and refinement of the student's 5 proposed corrections.
Independent Content & Validation	I proposed the specific corrective relationships based on the team's actual UI/workflow patterns. After confirmation, I manually applied all the fixes into UseCaseSpec (2).md and verified each field.

Session 9: Use-Case Diagram Correction — Diagram 1, User Features (Attempt 1)

Field	Content
Tool name	Claude (Anthropic)
Version	Sonnet 3.5 High
Platform	claude.ai (Web)
Access time	09/07/2026, 09:43
Purpose of use	Correct the Mermaid Use-Case Diagram ("Diagram 1 – User Features") in UCSPEC.md so it accurately reflects all 21 use cases specified in section 2.1 "User Features."
Prompts used	1. Hãy đọc thật kĩ file UCSPEC.md ... Tuy nhiên phần Diagram 1 đang còn nhiều sai sót và chưa thể hiện đầy đủ và đúng chiều các mối quan hệ giống như phần Use Case Spec của 2.1 User Features. Hãy sửa lại cho tôi sơ đồ Diagram 1 theo đúng với những Use Cases đã được specified... [Đính kèm: UCSPEC.md]
Which content was generated by AI	Corrected Diagram 1 (User Features), adding missing «extend» relationships, fixing relationship types, and removing unsupported «include» relationships.
Independent Content & Validation	The student read through the corrected Diagram 1 and checked it line by line against section 2.1 User Features to see if the changes matched what was written in the use case spec.

Session 10: Use-Case Diagram Correction — Diagram 1, User Features (Attempt 2, near-duplicate session)

Field	Content
Tool name	Claude (Anthropic)
Version	Sonnet 3.5 High
Platform	claude.ai (Web)
Access time	09/07/2026, 09:43

Field	Content
Purpose of use	Same task as Session 9, run in a second parallel session: correct Diagram 1 (User Features) in UCSPEC.md against section 2.1 as source of truth.
Prompts used	1. [Tương tự Prompt của Session 9]
Which content was generated by AI	Corrected Diagram 1 (User Features), identifying the same missing extend relationship and flagging contradictions, with system-level use cases explicitly excluded.
Independent Content & Validation	The student compared this output with the one from the earlier session and ended up keeping this session's result since it explained more clearly why system-level use cases should be left out.

Session 11: Use-Case Diagram Correction — Diagrams 1–3 and Full System Overview (all actor groups)

Field	Content
Tool name	Claude (Anthropic)
Version	<i>Sonnet 3.5 High</i>
Platform	claude.ai (Web)
Access time	09/07/2026, 15:09 – 21:07
Purpose of use	Correct the Mermaid Use-Case Diagrams for User Features (Diagram 1), Blood Center Staff Features (Diagram 2), and Hospital Features (Diagram 3) against their respective Use-Case Spec sections; restructure the System Features diagram and the Full System Overview Diagram.
Prompts used	1-3. [Yêu cầu sửa Diagram 1, 2, 3 dựa trên file UCSPEC tương ứng] 4-5. [Yêu cầu cải thiện bố cục trực quan cho biểu đồ System 4.1/4.2] 6. Hãy đọc thật kĩ file UseCaseSpec (2).md, hiện tại file có phần Full System Overview Diagram là chưa chính xác... [Đính kèm: UseCaseSpec (2).md, 01_UseCaseDiagram_slides_2015.pdf]
Which content was generated by AI	Corrected Diagram 1, 2, 3; a visually restructured System Features diagram; and a corrected Full System Overview Diagram using dashed-border styling to distinguish in-scope vs. cross-diagram-reference use cases.

Field	Content
Independent Content & Validation	I checked all four diagrams one by one against their matching UseCaseSpec sections, made sure the layout changes didn't lose content, and confirmed the styling matched the reference slide deck.

Session 12: Instructor Feedback Follow-Up — Extend Relationship Clarification & Final Diagram Rewrite

Field	Content
Tool name	Claude (Anthropic)
Version	<i>Sonnet 3.5 High</i>
Platform	claude.ai (Web)
Access time	22/07/2026, 20:26 – 24/07/2026, 23:51
Purpose of use	Interpret instructor feedback on extend relationships (navigation issue); get conceptual explanation and re-audit every extend relationship in the User Features diagram to correct them.
Prompts used	1. [Gửi feedback của thầy và hỏi cách hỏi lại thầy cũng như cách sửa cho hợp lý] 2. Giải thích và phân tích kĩ cho tôi thế nào mới đúng là extend đi? 3. Hãy chỉ cho tôi cách sửa cho toàn bộ các mối quan hệ extend. Nếu là Extend hãy ghi rõ Extension Point là ở đâu. Còn nếu xác định nó không phải mối quan hệ Extend thì hãy loại bỏ nó... [Đính kèm: UserUseCaseSpec.md]
Which content was generated by AI	Identification of ~33 problematic extend relationships; conceptual breakdown of valid extend relationships; and a full Keep/Remove/Convert-to-Include rewrite of every relationship with explicit Extension Points.
Independent Content & Validation	The student went through every one of the ~33 extend relationships using the 4-question test provided by AI, deciding case by case whether each should stay, be removed, or changed, and verified the final diagram match.

Session 13: Auth & Account Module – System Init, Zod Schema & QR Code Registration

Field	Content
Tool name	Gemini
Version	1.5 Pro / 2.0 Flash
Platform	gemini.google.com (Web)
Access time	23/07/2026 – 25/07/2026
Purpose of use	Analyze the initial authentication flow, understand Zod validation, convert manual registration into a CCCD QR code scanning implementation, and debug Mongoose/Swagger issues.
Prompts used	1-3. [Yêu cầu phân tích luồng, giải thích zod và sửa tính năng đăng ký QR code] 4-5. [Hỏi về việc decode QR ở FE hay BE và ảnh hưởng của các trường dữ liệu CCCD] 6-10. [Yêu cầu sửa model user/profile, giải thích lỗi code và database collection]
Which content was generated by AI	Explanation of the request lifecycle, guide on Zod schema inference, architectural advice on QR decoding latency, and payload extraction logic for the decoded QR string.
Independent Content & Validation	I used my own CCCD to scan and check actual plaintext data fields. I updated the Zod schema and Mongoose models myself, fixed the MongoDB Atlas connection, and tested via Swagger UI.

Session 14: Auth & Account Module – Email Verification, Password Reset & JWT Security

Field	Content
Tool name	Gemini
Version	1.5 Pro / 2.0 Flash
Platform	gemini.google.com (Web)
Access time	23/07/2026 – 25/07/2026

Field	Content
Purpose of use	Rework password reset and email verification flows, integrate Brevo email API, resolve token/OTP conflicts, and implement JWT-based logout logic.
Prompts used	1-4. [Yêu cầu sửa API forgot password, giải quyết xung đột OTP/Token và logic logout] 5-7. [Sửa lỗi Brevo IP, lỗi verify-email link và giải thích phương thức GET/POST] 8-10. [Yêu cầu sửa luồng reset password từ Token sang OTP 6 số theo Prototype FE và kiểm tra logic hết hạn]
Which content was generated by AI	Code to separate reset/verification tokens, explanation of stateless JWT logout, Brevo API integration setup, and a full 3-step rework of the reset password flow.
Independent Content & Validation	The student compared the Express version with an older Python version. The student set up the Brevo account and tested email delivery. Noticed and removed redundant OTP verification steps to match the frontend expectations.

Session 15: Auth & Account Module – Profile Management, Cloudinary & Dynamic Payloads

Field	Content
Tool name	Gemini
Version	1.5 Pro / 2.0 Flash
Platform	gemini.google.com (Web)
Access time	23/07/2026 – 25/07/2026
Purpose of use	Set default values for donor profiles, integrate Cloudinary for avatar uploads, fix MongoDB validation errors, and design dynamic PATCH payloads.
Prompts used	1-3. [Khởi tạo xp/level, avatar mặc định, bổ sung Cloudinary và hướng dẫn setup tài khoản] 4-6. [Sửa lỗi Mongo validation Error 121, bổ sung trường update địa chỉ/phone/email]

Field	Content
	7-9. [Yêu cầu xử lý update trường lựa chọn (optional) và thảo luận hướng tối ưu xử lý ảnh ở FE hay BE]
Which content was generated by AI	Code for default profile fields, Cloudinary and Multer setup guide, fix for MongoDB validation issues, and Zod schema patterns for partial updates.
Independent Content & Validation	The student set up the Cloudinary account, uploaded the default avatar, and configured environment variables. Tested the update profile feature in Swagger UI field by field.

Session 16: Auth & Account Module – GeoJSON Schema Refactoring

Field	Content
Tool name	Gemini
Version	1.5 Pro / 2.0 Flash
Platform	gemini.google.com (Web)
Access time	23/07/2026 – 25/07/2026
Purpose of use	Debug MongoDB geospatial errors, understand HTML5 geolocation vs database storage, and refactor the schema to separate display address from coordinates.
Prompts used	1. Sao tôi nhập input này để thay đổi thì nó báo lỗi này MongoServerError: Can't extract geo keys: {...} unknown GeoJSON type... 2. [Hỏi về việc lưu location tọa độ người dùng khi dùng Goong API] 3-5. [Yêu cầu tạo trường currentAddress mới thay cho location, hỏi về kiểu dữ liệu bsontype] 6. Hiện tại đây là response trả về khi tôi execute tính năng getprofile, hiện tại nó vẫn đang để trường location chứ không phải currentAddress, hãy giúp tôi sửa nó lại.
Which content was generated by AI	Explanation of MongoDB 2dsphere indexing, GeoJSON requirements, and HTML5 geolocation. Suggested schema changes for currentAddress and corresponding TypeScript/Service mapping logic.

Field	Content
Independent Content & Validation	The student reviewed the QR payload to confirm permanentAddress should remain a string. Applied schema changes via MongoDB Compass, updated code, and verified the API response via get/update endpoints.

B.2. Trần Minh Triết – Developer

Session 1: Use Case Specification – Campaign & Blood Center Management

Field	Content
Tool name	Claude (Anthropic)
Version	Sonnet 5
Platform	claude.ai (Web)
Access time	01/07/2026 – 05/07/2026
Purpose of use	Draft the complete Use Case Specification for system use cases (specifically focusing on Campaign and Blood Center Management) based on business requirements and UI screens, ensuring the output follows the required course format.
Prompts used	1. "Based on the website description, business requirements, and UI screens I provide, complete the Use Case Specification for all use cases in the required course format, including Basic Flow, Alternative Flows, Preconditions, Postconditions, Exception Flows, and Special Requirements where applicable."
Which content was generated by AI	Draft versions of the Use Case Specification (UCSPEC) for identified use cases, including actor interactions, preconditions, postconditions, and all flow types. Assisted in checking relationships («include», «extend») for consistency.
Independent Content & Validation	The student designed the original website structure, business requirements, and functional descriptions. After AI generated the drafts, the student reviewed each specification, corrected business logic errors, refined

Field	Content
	workflow details, and verified that all relationships accurately reflected the intended system behavior.

Session 2: UI Prototype Generation (Google Stitch)

Field	Content
Tool name	Stitch (Google)
Version	N/A
Platform	Google Stitch
Access time	15/06/2026 – 20/06/2026
Purpose of use	Quickly produce UI prototypes (wireframes/mockups) for the Campaign Management module to visualize the layout before moving to actual implementation.
Prompts used	1. "Generate a UI prototype for the Campaign Management screens based on the feature description."
Which content was generated by AI	UI prototypes (wireframes/mockups) for the Campaign Management module, including Campaign List, Create Campaign, and Campaign Details/Edit pages.
Independent Content & Validation	The student reviewed the AI-generated prototypes, manually adjusted the layout and data fields to match real business requirements (e.g., adding specific blood type demand fields), and ensured navigation flow was consistent with the approved Use Case Specifications.

Session 3: Spec-Driven Development (speckit) & Coding Implementation

Field	Content
Tool name	Claude (Anthropic)
Version	Sonnet 5 / Claude Code CLI
Platform	Claude Code

Field	Content
Access time	20/07/2026
Purpose of use	Draft technical prompts standardized for the Spec-Driven Development (speckit) workflow and assist in the implementation and code review of use cases BC-UC-01, 02, and 03.
Prompts used	1. "Write English prompts for speckit specify/plan/tasks/implement to deliver BC-UC-01, 02, 03, with a strict requirement to read the existing code structure and only add code, without modifying or deleting existing code."
Which content was generated by AI	A complete set of technical prompts for each stage of the speckit workflow (specify , plan , tasks , implement). Assisted in reviewing generated code, checking Zod schemas against Mongoose models, and detecting a registration conflict between the campaign and booking modules.
Independent Content & Validation	The student personally executed each prompt in the CLI, observed execution results (Swagger UI, MongoDB Compass), and identified a 500 error caused by duplicate model naming. The student required the AI to investigate the root cause but made the final decision on all database and source code changes.

B.3. Nguyễn Quốc Dương – Developer

Session 1: Feature Specification Generation (/specify)

Field	Content
Tool name	Gemini (Google Antigravity)
Version	3.6 Flash & 3.1 Pro
Platform	Antigravity IDE & CLI (Web/Desktop)
Access time	09:30, 20/07/2026
Purpose of use	Use the speckit skill to analyze requirements for Functional Group FG2: Donation Booking & Location (LL-UC-06 to LL-UC-10) and create a structured feature specification (spec.md).

Field	Content
Prompts used	1. "Use the speckit skill to analyze LL-UC-06 to LL-UC-10 (Donation Booking & Location) and create feature specification artifact (spec.md) for FG2."
Which content was generated by AI	Initial feature specification (<code>spec.md</code>); core entity attribute definitions (Appointment, E-Ticket, Screening Form, Campaign); primary user flow and error handling scenarios for location browsing and booking.
Independent Content & Validation	The student reviewed the generated specification against LifeLine requirements, refined specific business constraints (e.g., minimum 84-day interval between donations, cancellation grace period), and updated API contracts to align with team design guidelines.

Session 2: Architectural Planning & Task Breakdown (/plan & /tasks)

Field	Content
Tool name	Gemini (Google Antigravity)
Version	3.6 Flash & 3.1 Pro
Platform	Antigravity IDE & CLI
Access time	14:15, 20/07/2026
Purpose of use	Create a detailed architectural implementation plan (<code>plan.md</code>) and dependency-ordered tasks (<code>tasks.md</code>) for the TypeScript backend-core booking module.
Prompts used	1. "Use speckit-plan and speckit-tasks to create an architectural implementation plan (plan.md) and dependency-ordered tasks (tasks.md) for the TypeScript backend-core booking module."
Which content was generated by AI	Step-by-step implementation plan (<code>plan.md</code>); dependency-ordered task list (<code>tasks.md</code>) covering API endpoints, validation logic, and database integration.
Independent Content & Validation	The student ensured strict reuse of existing shared infrastructure (middleware, DB utilities), adjusted the module structure to match

Field	Content
	<code>src/backend-core/src/modules/booking</code> , and manually prioritized tasks starting from models and schemas.

Session 3: Data Models & Mongoose Schemas Preparation

Field	Content
Tool name	Gemini (Google Antigravity)
Version	<i>3.6 Flash & 3.1 Pro</i>
Platform	Antigravity IDE & CLI
Access time	10:00, 21/07/2026
Purpose of use	Design TypeScript Mongoose schemas and models for Appointments, E-Tickets, and Screening Forms.
Prompts used	1. "Design TypeScript Mongoose schemas and models for Appointment, ETicket, ScreeningForm, and ScreeningFormTemplate in <code>src/backend-core/src/modules/booking/models</code> ."
Which content was generated by AI	Mongoose model files (<code>appointment.model.ts</code> , <code>eticket.model.ts</code> , etc.); initial field definitions and enum values for appointment status.
Independent Content & Validation	The student migrated AI-suggested generic JS models to TypeScript Mongoose interfaces, integrated dynamic <code>ScreeningFormTemplate</code> logic, and added critical geospatial 2DSphere indexes on campaign/center coordinates for location discovery.

Session 4: Core Business Logic & Express TypeScript Implementation

Field	Content
Tool name	Gemini (Google Antigravity)
Version	<i>3.6 Flash & 3.1 Pro</i>
Platform	Antigravity IDE & CLI

Field	Content
Access time	15:45, 22/07/2026
Purpose of use	Implement Booking Service, Controller, Zod validation schemas, and Express routes for the booking lifecycle.
Prompts used	1. "Implement BookingService, BookingController, Zod request validation schemas, and Express routes for appointment creation, cancellation grace period, screening form submission, and location search."
Which content was generated by AI	Express routes, controller handlers, and service business logic; Zod validation schemas (create-appointment.schema.ts , etc.).
Independent Content & Validation	The student enforced exact questionnaire length validation in Zod, implemented specific cancellation grace period rules, and integrated the qrcode and cloudinary modules for E-Ticket generation.

Session 5: Automated Unit & Integration Testing

Field	Content
Tool name	Gemini (Google Antigravity)
Version	3.6 Flash & 3.1 Pro
Platform	Antigravity IDE & CLI
Access time	11:20, 23/07/2026
Purpose of use	Create Jest unit and integration tests covering happy paths, overbooking scenarios, and donor eligibility rules.
Prompts used	1. "Create Jest unit tests (booking.service.test.ts) and integration tests (booking.integration.test.ts) covering happy paths, slot overbooking, donor eligibility rules, and invalid form data."
Which content was generated by AI	Unit test suite and integration test suite files; Supertest assertions for API HTTP response codes.
Independent Content &	The student fixed mock database dependencies, resolved TypeScript type mismatches, authored additional edge-case tests (e.g., donor interval

Field	Content
Validation	eligibility checks), and verified the test pass rate locally.

Session 6: AI Usage Documentation & Report Alignment

Field	Content
Tool name	Gemini (Google Antigravity)
Version	3.6 Flash & 3.1 Pro
Platform	Antigravity IDE & CLI
Access time	22:30, 24/07/2026
Purpose of use	Finalize the AI Usage Notes report to align with the actual technology stack and course guidelines.
Prompts used	1. "Review the execution history prompt and edit the AI Usage report to align perfectly with the actual project technology stack, file structure, and course guidelines."
Which content was generated by AI	Standardized Markdown structure for the report; synchronized session logs.
Independent Content & Validation	The student audited all prompts and file paths against the actual repository logs, ensuring consistent branding and adherence to Academic Integrity standards.

B.4. Trần Đức Quý – FE/BE & UI/UX Developer

Session 1: Use-Case Specification & Design Documentation (Notification, AI Chatbot, Admin)

Field	Content
Tool name	Gemini
Version	1.5 Pro
Platform	gemini.google.com (Web)

Field	Content
Access time	July 2026 (Sprint 3)
Purpose of use	Draft the Use-Case Specifications for the Notification, AI Chatbot, and Admin features based on the Vision Document, and generate the initial DESIGN.md file for the Blood Center UI/UX guidelines.
Prompts used	1. "Dựa vào file vision doc, hãy viết Use Case Spec cho các phần Notification, AI Chatbot và Admin. Cần tuân thủ đúng định dạng mẫu của team. Sau đó, tạo file design.md để làm guideline UI cho phần Blood Center (BC)."
Which content was generated by AI	Markdown tables for the Use-Case Specifications (Basic Flows, Alternative Flows, Pre/Post-conditions) for Notification, Chatbot, and Admin. A structured DESIGN.md file detailing color palettes, typography, and component states for the Blood Center.
Independent Content & Validation	The student reviewed the generated alternative flows to ensure they accurately reflected system edge cases (e.g., unrecognized chatbot queries). The student also refined the DESIGN.md tokens to match existing global variables and the team's design system.

Session 2: UI/UX Prototyping for Blood Center Features

Field	Content
Tool name	Google Stitch
Version	Web App
Platform	Stitch Platform
Access time	July 2026 (Sprint 3)
Purpose of use	Generate high-fidelity UI prototypes for the Blood Center's Notification Hub and AI Chatbot interfaces based on the newly created specifications.
Prompts used	1. "Tạo UI prototype cho giao diện Blood Center, tập trung vào Notification hub và cửa sổ AI Chatbot. Yêu cầu thiết kế gọn gàng, trực quan và đồng nhất với design system đã cung cấp trước đó."

Field	Content
Which content was generated by AI	Interactive prototype screens for the Blood Center dashboard, displaying a slide-out notification drawer and a floating AI Chatbot widget.
Independent Content & Validation	The student evaluated the prototypes against the user flows in the UCSPEC. The student manually added "empty state" illustrations for the notification hub and adjusted the layout grid to prevent the chatbot window from overlapping data tables on smaller screens.

Session 3: Frontend & Backend Integration (Figma to Code via MCP)

Field	Content
Tool name	Antigravity
Version	Antigravity (with MCP enabled)
Platform	Antigravity IDE
Access time	July 2026 (Sprint 3)
Purpose of use	Extract UI code directly from Figma designs using MCP, and generate integration logic connecting the React frontend with the Express backend for Admin and Notification modules.
Prompts used	1. "Sử dụng MCP để trích xuất code HTML/CSS từ liên kết Figma này và chuyển đổi thành React component. Phân chia component rõ ràng, không sử dụng inline style. Đồng thời, viết script tích hợp gọi API backend cho màn hình Admin, bao gồm xử lý lỗi CORS và thêm trạng thái loading."
Which content was generated by AI	React component files (TSX) translated from Figma frames, Axios fetch wrappers, and standard error-handling boilerplate for backend communication.
Independent Content & Validation	The student manually refactored the component hierarchy to ensure modularity and reusability. The student discarded AI-suggested CORS middleware that conflicted with server configs and implemented API state management using custom team React hooks.

B.5. Trịnh Khánh Linh

Session 1: Use Case Specification – FG 1.5 & FG 1.6

Field	Content
Tool name	ChatGPT
Version	<i>GPT-5.5</i>
Platform	chatgpt.com (Web)
Access time	28/06/2026 - 03/07/2026
Purpose of use	Understand the UCSPEC format and draft initial specifications for FG 1.5 (Donation Impact & Tracking) and FG 1.6 (Community) .
Prompts used	1. "Dựa trên tài liệu đặc tả của LifeLine, soạn thảo Use Case Specification cho nhóm tính năng FG 1.5 (Donation Impact & Tracking) và FG 1.6 (Community) theo đúng format: ID, Actor, Preconditions, Basic Flow, Alternative Flows và Related Use Cases." 2. "Kiểm tra sự nhất quán giữa các Alternative Flows của tính năng vinh danh (Achievements) và theo dõi hành trình hiến máu (Donation Timeline) để đảm bảo không bị trùng lặp logic."
Which content was generated by AI	Draft versions of Use Case Specifications (UCSPEC) including actor interactions, structured flows for tracking impacts, and suggested relationships for the Community features.
Independent Content & Validation	The student reviewed every generated use case against the project scope, removed unnecessary steps in the Community module, modified flows to reflect the intended gamification logic in Impact Tracking, and manually finalized the include/extend relationships.

Session 2: UI Prototype Design – Impact & Community Modules

Field	Content
Tool name	Google Stitch
Version	<i>Web Platform</i>
Platform	stitch.withgoogle.com

Field	Content
Access time	28/06/2026 – 05/07/2026
Purpose of use	Generate UI prototypes for Donation Impact & Tracking (Timeline, Achievements, Donor Level) and the Community page, including alternative flow states.
Prompts used	1. "Tạo các trạng thái UI cho nhóm Donation Impact & Tracking (bao gồm Timeline, Achievements, Donor Level) dựa trên thiết kế profile mẫu. Đảm bảo giữ vững hệ thống Design Tokens (color, spacing)." 2. "Tạo thêm màn hình Alternative Flow cho trạng thái 'No history' (chưa có lịch sử hiến máu) và 'No achievements' cho nhóm Impact Tracking." 3. "Thiết kế trang Community (FG 1.6) như một module độc lập, hiển thị Fanpage chính thức, tin tức cộng đồng và bảng vinh danh người hiến máu."
Which content was generated by AI	High-fidelity UI mockups for the Donation Impact sub-tabs and the main Community navigation page; layout suggestions for empty and error states.
Independent Content & Validation	The student manually reorganized the Community feature from a sub-tab into a primary sidebar navigation item, adjusted data fields to match the finalized UCSPEC, and ensured visual consistency across all healthcare dashboard components.

Session 3: Design Extraction & Implementation Planning (Donor UI/UX)

Field	Content
Tool name	Claude / Gemini (via Antigravity IDE)
Version	<i>Claude 3.5 / Gemini 1.5 Pro</i>
Platform	Antigravity (with MCP enabled)
Access time	20-23/07/2026
Purpose of use	Configure Figma-to-IDE integration (MCP) to extract design specifications for the FG 1.5 & 1.6 modules and plan the React implementation.

Field	Content
Prompts used	1. "Sử dụng Antigravity IDE để trích xuất Design Tokens và Layout từ link Figma của màn hình Impact Tracking và Community." 2. "Với vai trò Senior Frontend Developer, hãy lập kế hoạch phân rã công việc (Task Breakdown) để code các React component cho module vinh danh (Achievements) và bảng tin cộng đồng." 3. "Phân tích đặc tả kỹ thuật từ Figma node data và đề xuất cấu trúc component tối ưu cho module FG 1.5 & 1.6."
Which content was generated by AI	Technical specifications extracted from Figma frames; step-by-step implementation plan and component architecture suggestions for the Impact & Community modules.
Independent Content & Validation	The student manually applied the IDE configurations and managed security tokens. The student reviewed the AI's task breakdown, isolated complex Figma frames to improve data extraction, and independently determined the final component hierarchy for the React repository.